

Running GLS

The use of static tools to reduce GLS time should be used before running zero-delay information, especially for linting. Static-timing analysis can provide information that is used to start GLS early in the flow.

There are many reasons for running GLS, some of which are given below:

1. To give confidence in verification of low-power structures, absent in RTL and added during synthesis. It is a probable method to catch multi-cycle paths if tests exercising these are available. Power estimation is done on the netlist for power numbers
2. To verify the power-up and reset operation of the design and to check if the design has any unintentional dependencies on initial conditions
3. To verify DFT structures absent in RTL and added during or after synthesis. Scan chains are generally inserted after the gate-level netlist has been created. Hence, GLS is often used to determine whether scan chains are correct. It is also required to simulate ATPG patterns. Tester pattern simulations are done on the gate-level netlist
4. To help reveal glitches on edge-sensitive signals due to combinational logic; using both worst- and best-case timing may be necessary
5. To check special logic circuits and design topology that may include feedback and/or initial state considerations or circuit tricks
6. To check if design works at the desired frequency with actual delays in place. It is a probable method to find out the need for synchronisers, if absent in design. It will cause X propagation on timing violation on that flop

GLS execution strategy

In highly-integrated products, it is not possible to run gate simulation for all system on chip (SoC) tests due to the simulation and debug time required. Therefore the vectors

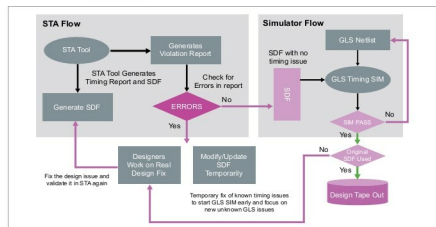


Fig. 2: Gate-level simulation and static-timing analysis flow

that are to be run in GLS have to be selected judiciously.

Possible candidates for such vectors are test cases involving initialisation and boot up, and all blocks of the design must have at least one test case for GLS, test cases checking clock source switching, cases checking clock frequency scaling, asynchronous paths in design, test cases that check entry/exit from different modes of design and dedicated tests for timing exceptions in the STA.

GLS targets the maximum desired operating frequency of the design. Some signals that are critical for GLS debug can be preserved during synthesis. A list of all synchroniser flops is generated using CDC tools.

Asynchronous paths where timing checks need to be turned off are analysed and a complete list of such flops is prepared, which also includes reset synchronisers. Timing checks are turned off on all such flops to avoid any redundant debugging, otherwise these will cause X corruption in GLS.

This work should ideally be done before the SDF arrives. It may happen that the names of the synchronisers in RTL and the netlist are different. All such flops should be updated as per the netlist. Also, correct standard cell libraries, correct models of analogue blocks and more should be picked for GLS.

Unit-delay GLS for test bench cleanup setup is done for unit delay

GLS and test cases that are planned to be run on gate level are run with this setup to clean the test bench. This is done because unit-delay simulations are relatively faster and all test bench/testcase-related issues can be resolved.

Running unit-delay GLS is recommended because one can catch most of the test bench/testcase issues before the arrival of SDF. After SDF arrives, focus should be more on finding the real design/timing issues. So one must make sure that the time does not get wasted in debugging test-case-related issues at that time.

GLS challenges

The challenge in GLS is X propagation debug. X corruption may be caused by a number of reasons such as timing violations, uninitialised memory and non-resettable flops. There generally are uninitialised flops in design which due to the architecture are guaranteed not to cause any problems. There is a need to find out all such flops in the design and initialise these to some random value (either zero or one) so as to mimic silicon. It gives a clear picture of how the design will behave at the desired frequency with actual delays in place.

Although GLS has its own set of challenges like set-up issues and long run time, among others, it is still very much a part of the sign-off process. **EFY**